

WASP-43b

R-1212

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_200405 · created 2026-07-31T07:54:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458944.6469 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.049
Transit depth (Rp/Rs)^2	1.89 +/- 1.34 [%]
Orbital Inclination (inc)	79.08 +/- 2.83
Airmass coefficient 1 (a1)	1.102 +/- 0.015
Airmass coefficient 2 (a2)	-0.074 +/- 0.056
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	#2 - [231, 78]
Optimal Method	PSF photometry
Transit Duration (day)	0.028 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.137 ± 0.049 vs 0.1594 (deviation 0.42σ)
Duration fit vs published	0.028 d vs 0.0512 d (deviation 1.41σ)
Mid-time O-C	0.8 min
Depth SNR	2.5
Residual scatter	1.32 %

Light curve

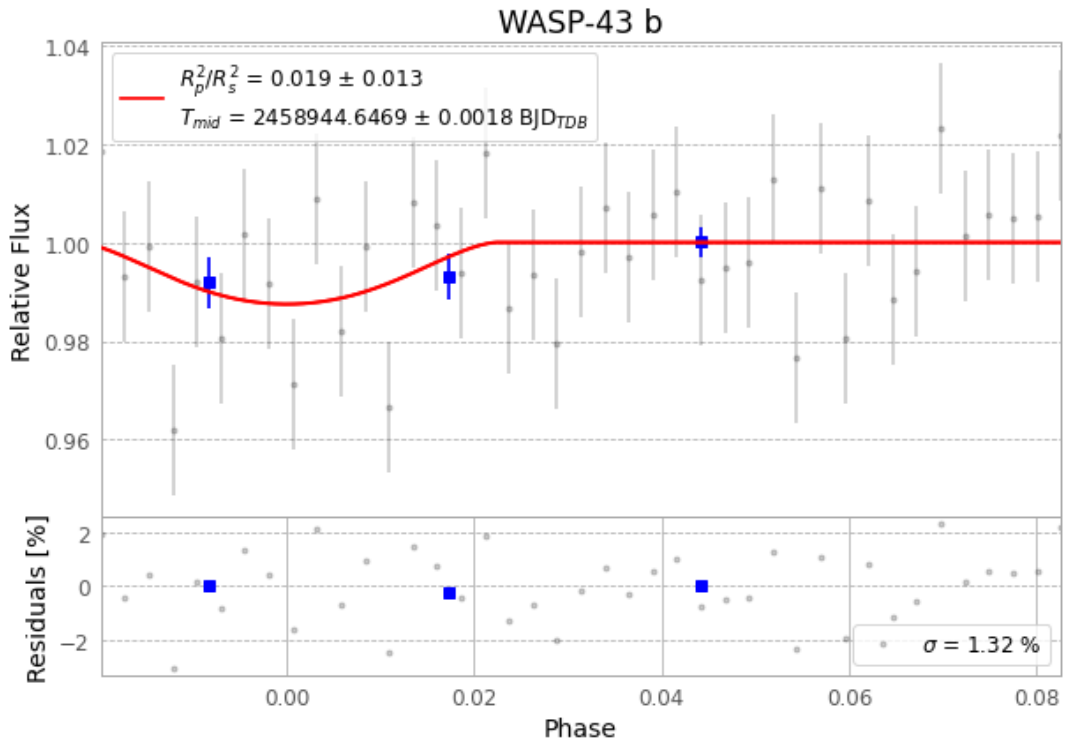


Figure 1 — FinalLightCurve_WASP-43 b_2020-04-04.png (SHA-256 9b250ce0087dc212...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 88cf97173b83c17f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

41 FITS frames (27 MB), WASP-43200405030012.FITS → WASP-43200405050012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200405.log (ee1042b302a3...), FinalLightCurve_WASP-43 b_2020-04-04.png (9b250ce0087d...), FinalLightCurve_WASP-43 b_2020-04-04.csv (35aef14ff677...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:54Z.